

WASP-12b

R-0565

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_170124 · created 2026-07-23T20:27:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457777.77874 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.045
Transit depth (Rp/Rs)^2	2.4 +/- 1.4 [%]
Orbital Inclination (inc)	77.07 +/- 2.96
Airmass coefficient 1 (a1)	0.7 +/- 0.29
Airmass coefficient 2 (a2)	0.22 +/- 0.18
Scatter in the residuals of the lightcurve fit is	37.64 %
Best Comparison Star	#2 - [109, 185]
Optimal Aperture	10.42
Optimal Annulus	22.57
Transit Duration (day)	0.0434 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.045 vs 0.1178 (deviation 0.58σ)
Duration fit vs published	0.0434 d vs 0.125 d (deviation 7.34σ)
Mid-time O–C	-0.9 min
Depth SNR	2.4
Residual scatter	37.64 %

Light curve

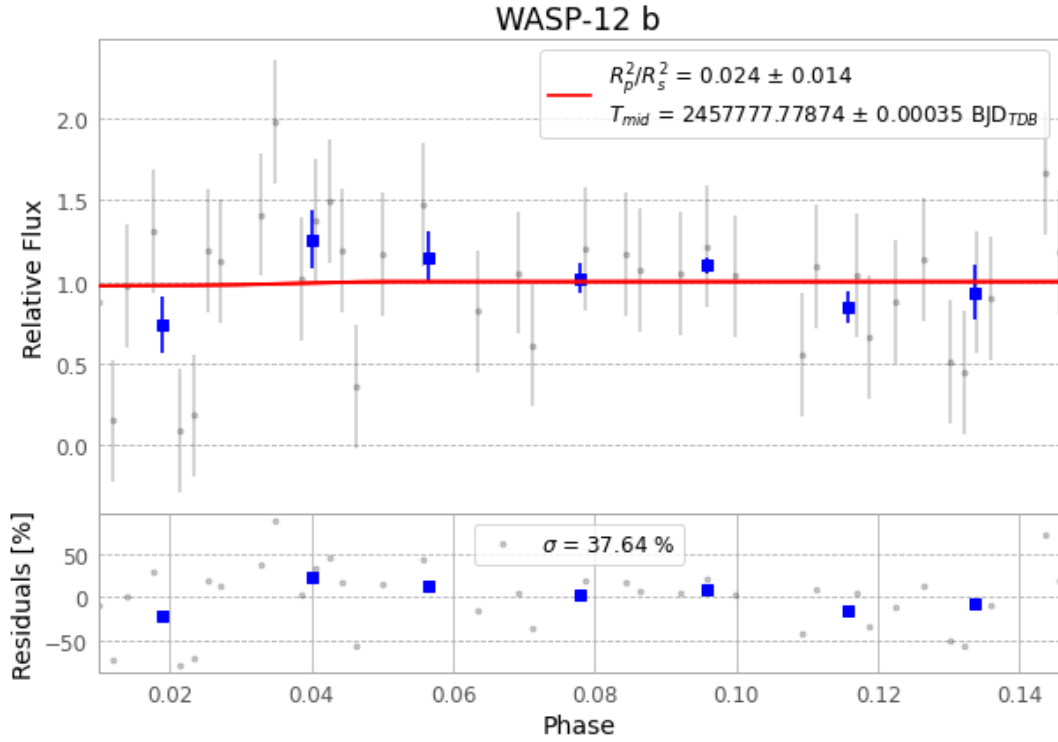


Figure 1 — FinalLightCurve_WASP-12 b_2017-01-23.png (SHA-256 595a9ec87136ce1b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [281, 147], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a818bef3d0619e55...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

72 FITS frames (47 MB), WASP-12170124064812.FITS → WASP-12170124102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-170124.log (7a9edae9b2ea...), FinalLightCurve_WASP-12 b_2017-01-23.png (595a9ec87136...), FinalLightCurve_WASP-12 b_2017-01-23.csv (ed7effd13997...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 20:27Z.